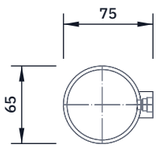
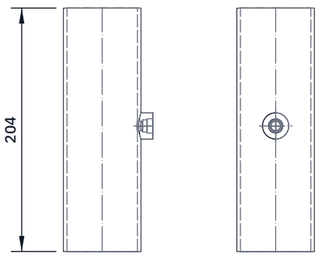
REV	DESCRIPTION	DATE	BY
P1	Making sketch for the prototype build plan	2026-10-01	AC



TOP VIEW

Scale 1:5; looking down (along -Z)

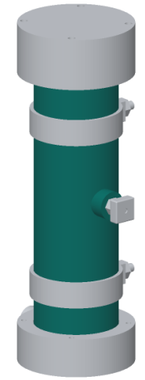


FRONT VIEW

Scale 1:5; looking at the front (along +Y)

RIGHT VIEW

Scale 1:5; looking at the right side (along -X)



WHERE IT GOES

This part in colour, its neighbours in grey

HOW TO MAKE IT AND HOW IT FITS

Cut 204 long, ends faced square and deburred (200 between caps plus 2 into each cap's seat). Keep the outside smooth near the ends: the O-rings seal on the end faces.

Boss: 22 dia x 14 long from 316 bar, one end bored to the tube's curve. TIG weld it on, centred 105 up from the bottom end, with argon purge inside; keep weld bead off the bore.

After welding: drill 8 through the boss and the wall; from the boss face cut an M12 x 1 thread 5 deep, then a 10.2 dia window seat 3.5 deeper, flat bottomed.

Pickle and passivate. The bore must be free of weld spatter so the liner slides in.

Check: the liner slides through by hand; the boss axis is square to the tube axis.

LUMAFLOW github.com/BoujeeEnjinia1701/lumaflow

LumaFlow reactor tube with sensor boss: making sketch

DRAWING NO.	REV	SHEET	SIZE
LMF-DWG-106	P1	1 of 1	ANSI B

SCALE	UNITS	PROJECTION	DATE	DRAWN
1:5	mm		2026-10-01	Amish Chadha

MATERIAL / NOTES

316 stainless tube 65 x 3; 316 bar 25 mm for the boss; machine shop

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